

**Gisma
University
of Applied
Sciences**

Assessment Submission Form

Student Number	GH1039590
Assessment Title	Reducing Wait Times & No-Shows in Outpatient Healthcare: A Statistical Analysis
Module Code	B105A
Module Title	Applied Statistical Modelling (SS0626)
Module Tutor	Prof. William Baker Morrison
Date Submitted	18/09/2026
GitHub link	https://github.com/Amnasayeeq/B105-ASM-Healthcare-NoShow-Analysis
Kaggle Dataset Link	https://www.kaggle.com/datasets/joniarroba/noshowappointments?resource=download

Declaration of Authorship

I declare that all material in this assessment is my own work except where there is a clear acknowledgement and appropriate reference to the work of others.

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Signed By AMNA

Date 18/09/2026

1. Business Problem

Most outpatient clinics have to set more appointments than they can to keep up with as a certain percentage of patients fail to appear for their appointment. Each missed appointment is an empty spot: The clinician has no income from the appointment, the room is not occupied, and the patient is still waiting for an available appointment. This adds up to actual costs, both economic and in terms of the number of patients available for care, over a large scheduling system. This project is not a "no show" project. Instead of a blanket policy, if a clinic knows, in advance, that a given appointment is at high risk of a no show, they can take action on that appointment, such as a reminder call, a courtesy SMS or a follow up phone call, instead of leaving it as is.

Main business question: What are the factors that are correlated with a patient missing their appointment and what proportion of this risk is explainable or actionable?

2. Business Questions and Hypotheses:

1. What percentage of the scheduled appointments is not attended?
2. Is there a difference in the likelihood of a no show appointment depending on the difference in days between booking the appointment and the appointment day (waiting time)?
3. Do SMS reminders actually have an impact on reducing no shows?
4. Is waiting time systematically different for each age group of patients?
5. How well can the combination of these factors predict a no show, and which of them is best able to do so?

2.1. Hypotheses & tests for evaluation

	Hypothesis	Test
H1	Mean waiting time differs between patients who show up and patients who no-show	Two-sample- T-Test
H2	SMS reminder receipt/No-show status is correlated	Chi-square Test
H3	There are differences between the mean waiting time for different age groups (Child / Teen / Adult / Senior).	ANOVA (One way)
H4	Waiting time, SMS receipt, age, and scholarship status and chronic conditions.	Multiple logistic regression

3. Dataset

The data was compiled from "Medical Appointment No Shows" (Kaggle), which consists of ~110,000 outpatient appointments made in public health clinics in Vitoria, Brazil (2016). Basic data about each scheduled appointment, including the patient's demographics and a few flags for chronic conditions, the result of sending an SMS reminder, and the result label No-show (Yes/No).

3.1. Dataset Dictionary

01 - PatientId

- Identification of a patient

02 - AppointmentID

- Identification of each appointment

03 - Gender

- Male or Female . Female is the greater proportion, woman takes way more care of they health in comparison to man.

04 - DataMarcacaoConsulta

- The day of the actual appointment, when they have to visit the doctor.

05 - DataAgendamento

- The day someone called or registered the appointment, this is before appointment of course.

06 - Age

- How old is the patient.

07 - Neighbourhood

- Where the appointment takes place.

08 - Scholarship

- True or False . Observation, this is a broad topic, consider reading this article https://en.wikipedia.org/wiki/Bolsa_Fam%C3%ADlia

09 - Hipertension

- True or False

10 - Diabetes

- True or False

Alcoholism

- True or False

Handcap

- True or False

SMS_received

- 1 or more messages sent to the patient.

No-show

- True or False.

Inspiration

What if that possible to predict someone to no-show an appointment?

4. Data Preparation, Sampling and Cleaning

In order to analyze the raw export, it is required to have three features:

The dates had to be parsed, some of the known data-entry errors had to be removed, and the categorical variables had to be typed correctly. Tests to perform to clean and the rationale for the tests.

```
1 #Data cleaning
2
3 #Question: Business problem; outpatient clinics losing capacity when patients miss scheduled appointments
4 # using dataset here from kaggle
5 # install.packages("tidyverse")
6 library(tidyverse)
7 raw <- read.csv("data/appointments_raw.csv", stringsAsFactors = FALSE)
8
9 cat(" Raw data structure \n")
10 str(raw)
11 cat("\n Missing values per column \n")
12 print(colSums(is.na(raw)))
13 cat("\nDuplicate AppointmentIDs:", sum(duplicated(raw$AppointmentID)), "\n")
14
15 #parse dates
16 raw$ScheduledDay <- as.Date(substr(raw$ScheduledDay, 1, 10))
17 raw$AppointmentDay <- as.Date(substr(raw$AppointmentDay, 1, 10))
18 raw$WaitingDays <- as.numeric(raw$AppointmentDay - raw$ScheduledDay)
19
20 #data-quality problems
21 n_start <- nrow(raw)
22
23 n_negative_age <- sum(raw$Age < 0)
24 n_negative_wait <- sum(raw$WaitingDays < 0)
25
26 cat("\n Data-quality issues found \n")
27 cat("Rows with Age < 0      :", n_negative_age, "(data-entry errors, dropped)\n")
28 cat("Rows with negative wait :", n_negative_wait, "(AppointmentDay before ScheduledDay, dropped)\n")
29
30 clean <- raw %>%
31   filter(Age >= 0, WaitingDays >= 0) %>%
32   distinct(AppointmentID, .keep_all = TRUE)
33
34 cat("\nRows removed in cleaning :", n_start - nrow(clean), "\n")
35 cat("Rows remaining          :", nrow(clean), "\n")
36
37 #conversions
38 clean <- clean %>%
39   mutate(
40     Gender      = factor(Gender, levels = c("F", "M")),
41     Scholarship = factor(Scholarship, levels = c(0, 1), labels = c("No", "Yes")),
42     Hipertension = factor(Hipertension, levels = c(0, 1), labels = c("No", "Yes")),
43     Diabetes     = factor(Diabetes, levels = c(0, 1), labels = c("No", "Yes")),
44     Alcoholism   = factor(Alcoholism, levels = c(0, 1), labels = c("No", "Yes")),
45     Handcap      = factor(ifelse(Handcap > 0, 1, 0), levels = c(0, 1), labels = c("No", "Yes")),
46     SMS_received = factor(SMS_received, levels = c(0, 1), labels = c("No", "Yes")),
47     `No.show`    = factor(`No.show`, levels = c("No", "Yes")),
48     Neighbourhood = factor(Neighbourhood),
49     AgeGroup     = cut(Age,
50                       breaks = c(-1, 12, 19, 59, 200),
51                       labels = c("Child", "Teen", "Adult", "Senior"))
52   )
53
54 write.csv(clean, "data/appointments_clean.csv", row.names = FALSE)
55 cat("\nCleaned dataset written to data/appointments_clean.csv\n")
```

Result

```
Data-quality issues found
> cat("Rows with Age < 0          :", n_negative_age, "(data-entry errors, dropped)\n")
Rows with Age < 0          : 1 (data-entry errors, dropped)
> cat("Rows with negative wait   :", n_negative_wait, "(AppointmentDay before ScheduledDay, dropped)\n")
Rows with negative wait   : 5 (AppointmentDay before ScheduledDay, dropped)
>
> clean <- raw %>%
+   filter(Age >= 0, WaitingDays >= 0) %>%
+   distinct(AppointmentID, .keep_all = TRUE)
>
> cat("\nRows removed in cleaning :", n_start - nrow(clean), "\n")

Rows removed in cleaning : 6
> cat("Rows remaining          :", nrow(clean), "\n")
Rows remaining          : 110521
```

Tests to perform to clean and the rationale for the tests.

- Rows with negative Age are errors in entering the data (should not be negative) and are dropped, not imputed, because there is no way to get the true value back.
- Logically impossible rows those with appointment dates prior to the scheduling date, are also eliminated. There is a logic to remove duplicate AppointmentIDs so that there is only one row per appointment.
- All binary flags are converted to factors (Scholarship, Hypertension, Diabetes, Alcoholism, Handicap, SMS_received), as is the outcome, since R's modelling functions work with factors rather than numbers.
- Age is then broken down into AgeGroup (Child / Teen / Adult / Senior) for group comparisons in H3.

5. Exploratory Data Analysis

5.1. Descriptive statistics

```

1 #EDA for Descriptive statistics
2 library(tidyverse)
3
4 data <- read.csv("data/appointments_clean.csv")
5
6 #numeric summary
7 #overall & split by outcome
8 numeric_summary <- data %>%
9   summarise(
10     Variable = "Age",
11     Mean = mean(Age), Median = median(Age), SD = sd(Age),
12     Min = min(Age), Max = max(Age)
13   ) %>%
14   bind_rows(
15     data %>% summarise(
16       Variable = "WaitingDays",
17       Mean = mean(WaitingDays), Median = median(WaitingDays), SD = sd(WaitingDays),
18       Min = min(WaitingDays), Max = max(WaitingDays)
19     )
20   )
21
22 cat(" Overall numeric summary \n")
23 print(numeric_summary)
24
25 waitingdays_by_outcome <- data %>%
26   group_by(No.show) %>%
27   summarise(
28     n = n(),
29     Mean_WaitingDays = mean(WaitingDays),
30     Median_WaitingDays = median(WaitingDays),
31     SD_WaitingDays = sd(WaitingDays)
32   )
33
34 cat("\n Waiting days by outcome \n")
35 print(waitingdays_by_outcome)
36
37 #frequency tables
38 noshow_rate_by_sms <- data %>% count(SMS_received, No.show) %>% group_by(SMS_received) %>% mutate(pct = round(100 * n / sum(n), 1))
39 noshow_rate_by_agegroup <- data %>% count(AgeGroup, No.show) %>% group_by(AgeGroup) %>% mutate(pct = round(100 * n / sum(n), 1))
40 noshow_rate_by_scholarship <- data %>% count(Scholarship, No.show) %>% group_by(Scholarship) %>% mutate(pct = round(100 * n / sum(n), 1))
41
42 cat("\n No-show rate (%) by SMS received \n")
43 print(noshow_rate_by_sms)
44 cat("\n No-show rate (%) by age group \n")
45 print(noshow_rate_by_agegroup)
46 cat("\n No-show rate (%) by scholarship status \n")
47 print(noshow_rate_by_scholarship)
48
49 dir.create("Results/descriptive-stats-results", showWarnings = FALSE, recursive = TRUE)
50 write.csv(numeric_summary, "Results/descriptive-stats-results/numeric_summary.csv", row.names = FALSE)
51 write.csv(waitingdays_by_outcome, "Results/descriptive-stats-results/waitingdays_by_outcome.csv", row.names = FALSE)
52 write.csv(noshow_rate_by_sms, "Results/descriptive-stats-results/noshow_rate_by_sms.csv", row.names = FALSE)
53 write.csv(noshow_rate_by_agegroup, "Results/descriptive-stats-results/noshow_rate_by_agegroup.csv", row.names = FALSE)
54 write.csv(noshow_rate_by_scholarship, "Results/descriptive-stats-results/noshow_rate_by_scholarship.csv", row.names = FALSE)
55
56 cat("\nDescriptive statistics saved to Results/descriptive-stats-results/\n")

```

Result

```

No-show rate (%) by SMS received
> print(noshow_rate_by_sms)
# A tibble: 4 x 4
# Groups:   SMS_received [2]
  SMS_received No.show    n  pct
  <chr>        <chr> <int> <dbl>
1 No          No      62509 83.3
2 No          Yes      12530 16.7
3 Yes         No      25698 72.4
4 Yes         Yes      9784 27.6
> cat("\n No-show rate (%) by age group \n")

```

```

No-show rate (%) by age group
> print(noshow_rate_by_agegroup)
# A tibble: 8 × 4
# Groups:   AgeGroup [4]
  AgeGroup No.show    n  pct
  <chr>    <chr>  <int> <dbl>
1 Adult   No      46606 79.1
2 Adult   Yes     12334 20.9
3 Child   No      16729 79.5
4 Child   Yes      4306 20.5
5 Senior  No      17931 84.7
6 Senior  Yes      3241 15.3
7 Teen    No       6941 74
8 Teen    Yes      2433 26
> cat("\n No-show rate (%) by scholarship status \n")

```

```

No-show rate (%) by scholarship status
> print(noshow_rate_by_scholarship)
# A tibble: 4 × 4
# Groups:   Scholarship [2]
  Scholarship No.show    n  pct
  <chr>       <chr>  <int> <dbl>
1 No         No      79924 80.2
2 No         Yes     19736 19.8
3 Yes        No      8283 76.3
4 Yes        Yes      2578 23.7

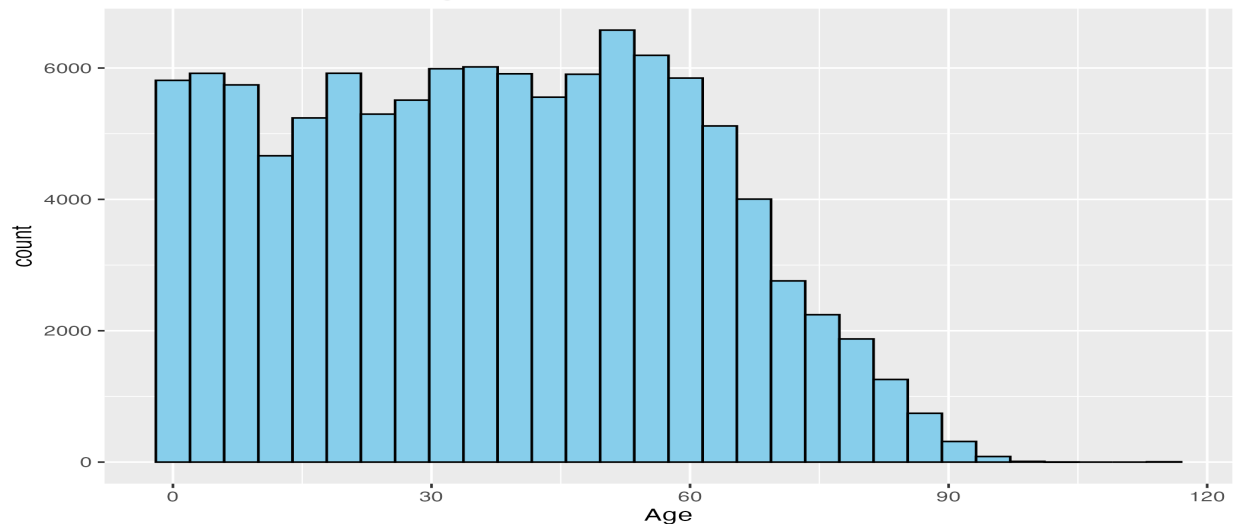
```

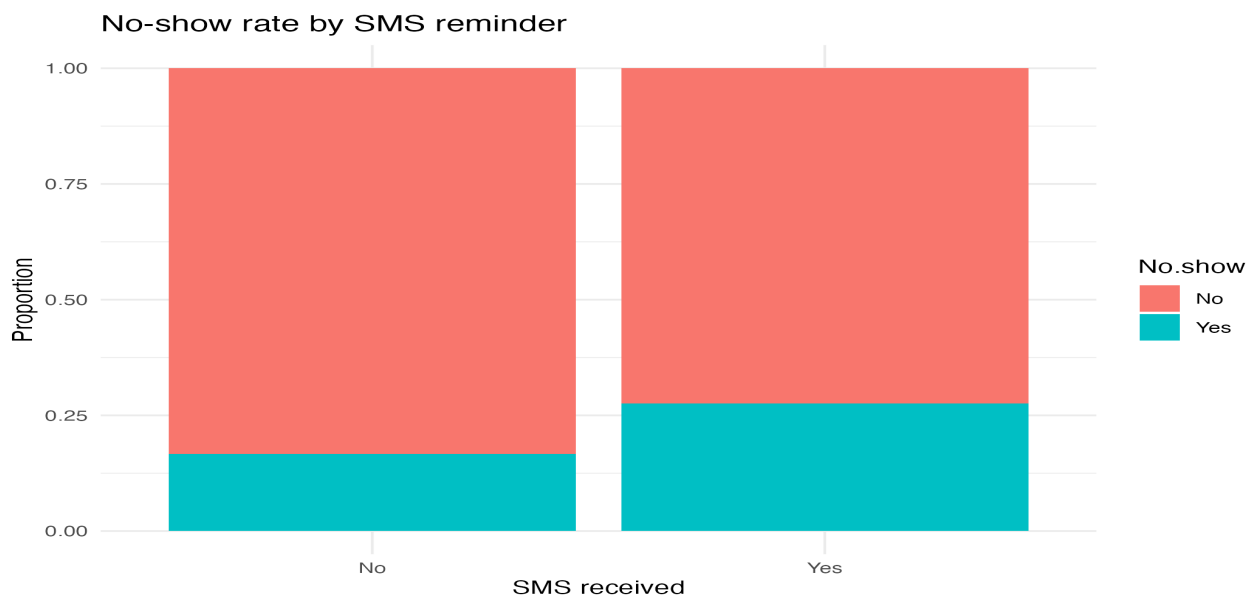
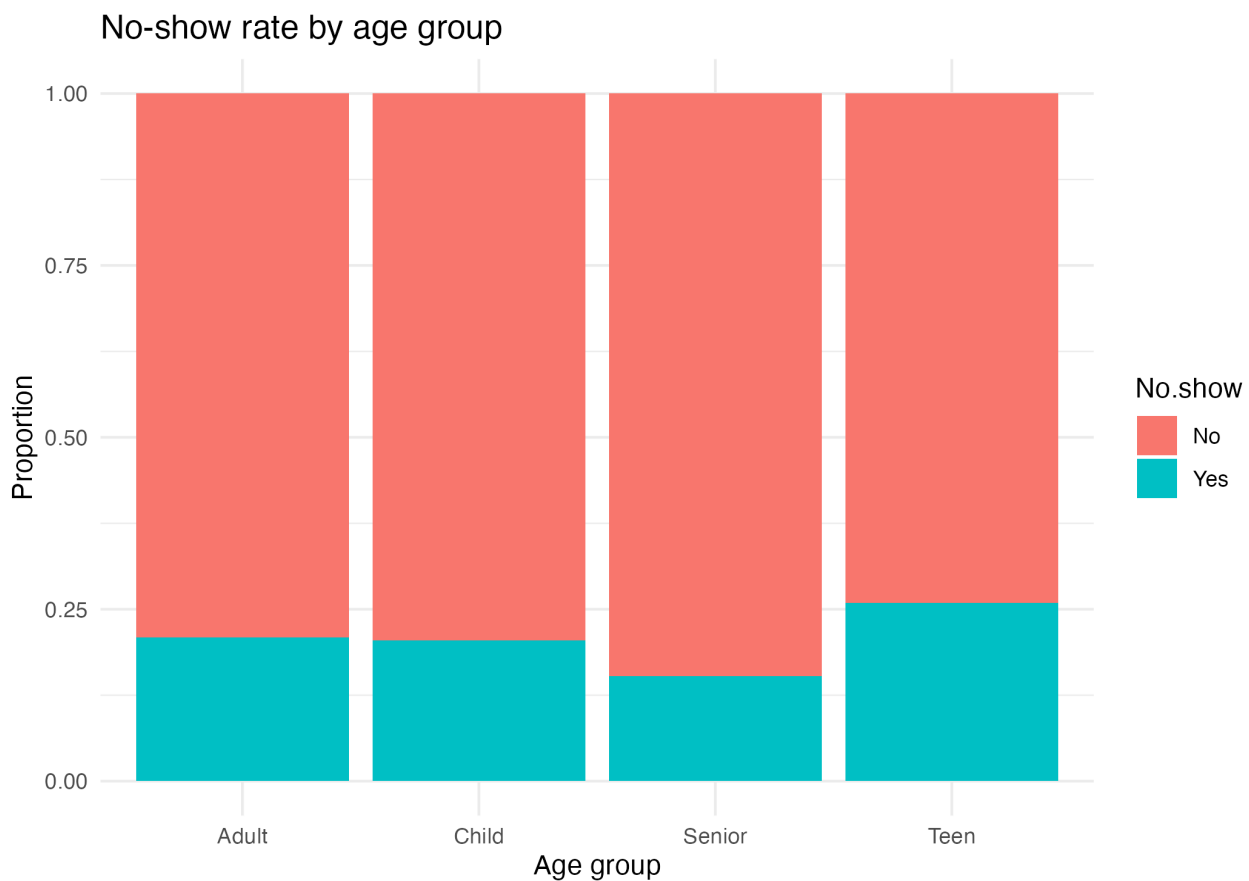
From all of the runs, the no-show rate for this one is 14.7%, this means that approximately 1 in 7 appointments is not attended. The waiting time is right skewed, the majority of patients have been seen in a couple of weeks after booking, but there is a long tail of bookings that wait for a much longer time, a point that suggests that a mean-based test should be subject to a careful check for normality.

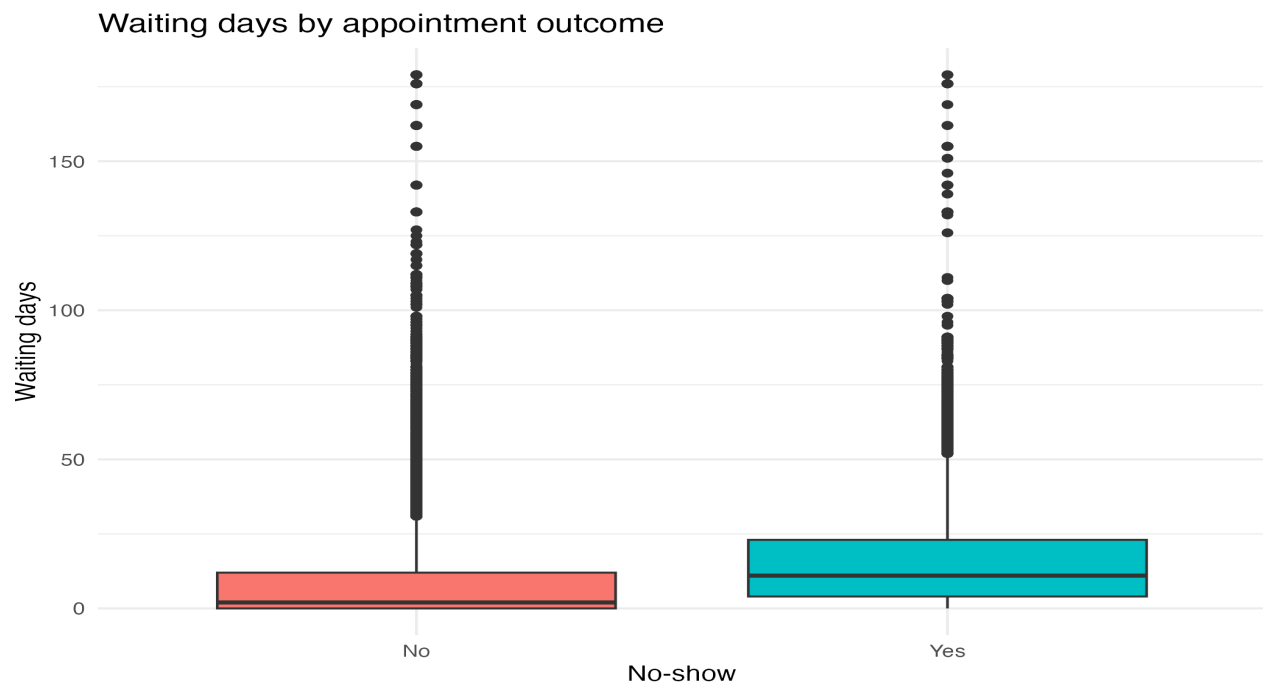
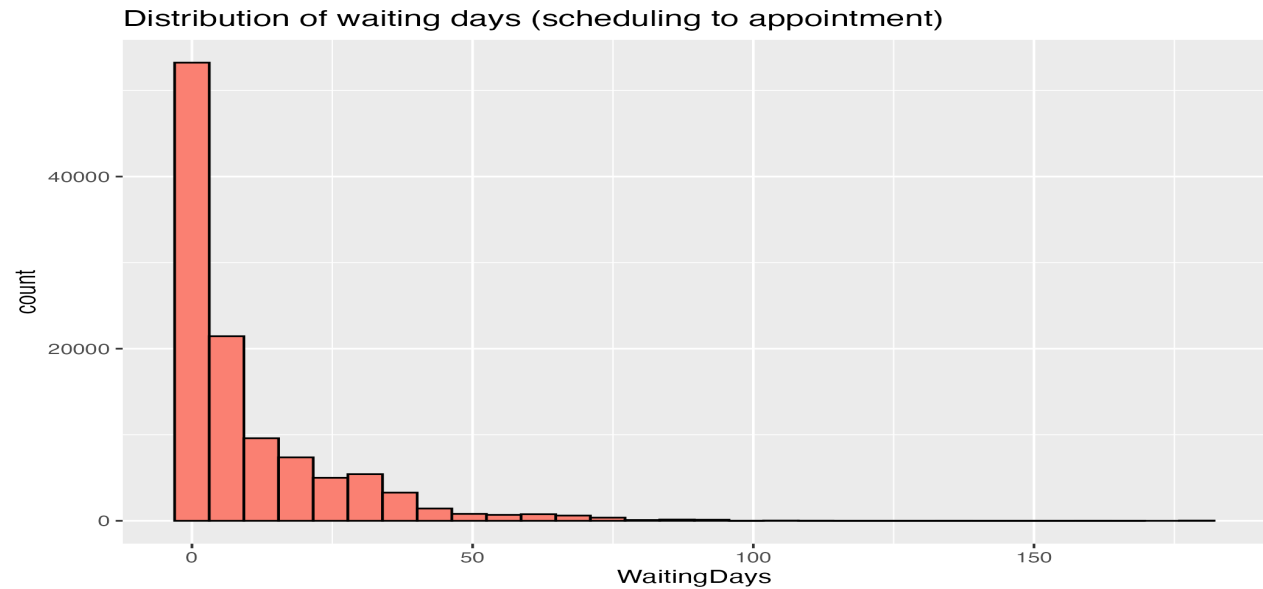
5.2. Visual exploration

```
1 #Exploratory Data Analysis
2
3 #the cleaned appointments dataset
4
5 library(tidyverse)
6
7 data <- read.csv("data/appointments_clean.csv")
8
9 cat(" No-show overall rate \n")
10 print(prop.table(table(data$No.show)))
11
12 #Distributions
13 pdf("Results/EDA-results/histograms.pdf")
14
15 ggplot(data, aes(x = Age)) +
16   geom_histogram(bins = 30, fill = "skyblue", color = "black") +
17   ggtitle("Distribution of patient age") -> p1
18 print(p1)
19
20 ggplot(data, aes(x = WaitingDays)) +
21   geom_histogram(bins = 30, fill = "salmon", color = "black") +
22   ggtitle("Distribution of waiting days (scheduling to appointment)") -> p2
23 print(p2)
24
25 dev.off()
26
27 ggsave("Results/EDA-results/age_distribution.png", plot = p1, width = 7, height = 5)
28 ggsave("Results/EDA-results/waitingdays_distribution.png", plot = p2, width = 7, height = 5)
29
30 #no-shows;
31 #SMS reminders
32 sms_plot <- data %>%
33   count(SMS_received, No.show) %>%
34   group_by(SMS_received) %>%
35   mutate(pct = n / sum(n)) %>%
36   ggplot(aes(x = SMS_received, y = pct, fill = No.show)) +
37   geom_col(position = "fill") +
38   labs(title = "No-show rate by SMS reminder", x = "SMS received", y = "Proportion") +
39   theme_minimal()
40 ggsave("Results/EDA-results/noshow_by_sms.png", plot = sms_plot, width = 7, height = 5)
41 print(sms_plot)
42
43 #age group
44 age_plot <- data %>%
45   count(AgeGroup, No.show) %>%
46   group_by(AgeGroup) %>%
47   mutate(pct = n / sum(n)) %>%
48   ggplot(aes(x = AgeGroup, y = pct, fill = No.show)) +
49   geom_col(position = "fill") +
50   labs(title = "No-show rate by age group", x = "Age group", y = "Proportion") +
51   theme_minimal()
52 ggsave("Results/EDA-results/noshow_by_agegroup.png", plot = age_plot, width = 7, height = 5)
53 print(age_plot)
54
55 #no-show vs waiting days
56 wait_box <- ggplot(data, aes(x = No.show, y = WaitingDays, fill = No.show)) +
57   geom_boxplot() +
58   labs(title = "Waiting days by appointment outcome", x = "No-show", y = "Waiting days") +
59   theme_minimal() + theme(legend.position = "none")
60 ggsave("Results/EDA-results/waitingdays_by_noshow_boxplot.png", plot = wait_box, width = 7, height = 5)
61 print(wait_box)
62
63 cat("\nEDA plots saved to Results/EDA-results\n")
```

Distribution of patient age







Before any formal tests, two patterns are observed:

First, the no-show patients seem to have slightly longer wait times on average investigate if these patterns are significant.

Second, no-show rate is clearly reduced among patients who were reminded by SMS

The no-show patients seem to have slightly longer wait times on average investigate if these patterns are significant.

6. Inferential Statistics

6.1. H1 — Waiting time: show vs. no-show

H0: The mean waiting time for the show group and the no-show group is equal.

H1: means are not equal.

```
1 #T-Test; waiting days
2 # Hypothesis: Is there a statistically significant difference in mean waiting time (the days between booking and appointment)
3 #b/w patients who show up and patients who no-show?
4 #H0: mu_show = mu_noshow
5 #H1: mu_show != mu_noshow
6
7 library(tidyverse)
8 library(car)
9
10 data <- read.csv("data/appointments_clean.csv") %>%
11   mutate(No.show = factor(No.show, levels = c("No", "Yes")))
12
13 show_wait <- data$WaitingDays[data$No.show == "No"]
14 noshow_wait <- data$WaitingDays[data$No.show == "Yes"]
15
16 #assumption 1: Normality
17 #the 5000 observations so we sample where its needed
18 cat("Shapiro-Wilk normality test (show group)\n")
19 print(shapiro.test(sample(show_wait, min(5000, length(show_wait)))))
20 cat("\n Shapiro-Wilk normality test (no-show group)\n")
21 print(shapiro.test(sample(noshow_wait, min(5000, length(noshow_wait)))))
22 #waiting days is right skewed
23 #expecting both p-values below 0.05, normality violated
24
25 #Assumption 2: Homogeneity of variance
26
27 cat("\n Levene's test for equal variances \n")
28 levene_result <- leveneTest(WaitingDays ~ No.show, data = data)
29 print(levene_result)
30
31 #Decision
32
33 #as the data is skewed it automatically fails the normality test
34 #therefore the groups are independent & we can't trust normality now we're just running both a&b
35 #(a) - Welch's t-test handles unequal variance & samples big
36 #(b) - Wilcoxon rank-sum test as a backup since the data is not normal
37 cat("\n Welch Two Sample t-test (primary) \n")
38 t_result <- t.test(WaitingDays ~ No.show, data = data)
39 #this is Welch by default
40 print(t_result)
41
42 cat("\n Wilcoxon rank-sum test (non-parametric confirmatory check) \n")
43 wilcox_result <- wilcox.test(WaitingDays ~ No.show, data = data)
44 print(wilcox_result)
45
46 #Visualisation no show waiting days
47 box_plot <- ggplot(data, aes(x = No.show, y = WaitingDays, fill = No.show)) +
48   geom_boxplot() +
49   labs(title = "Waiting days: show vs no-show",
50        subtitle = paste("Welch t-test p =", signif(t_result$p.value, 3),
51                          " | Wilcoxon p =", signif(wilcox_result$p.value, 3)),
52        x = "No-show", y = "Waiting days (days)") +
53   theme_minimal() + theme(legend.position = "none")
54
55 dir.create("Results/t-test-results", showWarnings = FALSE, recursive = TRUE)
56 ggsave("Results/t-test-results/waitingdays_boxplot.png", plot = box_plot, width = 7, height = 5)
57 print(box_plot)
58
59 density_plot <- ggplot(data, aes(x = WaitingDays, fill = No.show)) +
60   geom_density(alpha = 0.5) +
61   labs(title = "Distribution of waiting days by outcome") +
62   theme_minimal()
63 ggsave("Results/t-test-results/waitingdays_density.png", plot = density_plot, width = 7, height = 5)
64
65 cat("\nResults saved to Results/t-test-results\n")
66
```

Assumption checks

```
Shapiro-Wilk normality test (show group)
> print(shapiro.test(sample(show_wait, min(5000, length(show_wait)))))

      Shapiro-Wilk normality test

data:  sample(show_wait, min(5000, length(show_wait)))
W = 0.62554, p-value < 2.2e-16

> cat("\n Shapiro-Wilk normality test (no-show group)\n")

      Shapiro-Wilk normality test (no-show group)
> print(shapiro.test(sample(noshow_wait, min(5000, length(noshow_wait)))))

      Shapiro-Wilk normality test

data:  sample(noshow_wait, min(5000, length(noshow_wait)))
W = 0.82189, p-value < 2.2e-16

Levene's test for equal variances
> levene_result <- leveneTest(WaitingDays ~ No.show, data = data)
> print(levene_result)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group   1  1076.6 < 2.2e-16 ***
      110519
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

The data do not meet the equal variance assumption, either for Levene's test ($p < 0.001$) main reason why a Student's t-test is not appropriate here is that both of its assumptions are not met here, so the Student's t-test is not used as the primary test, instead a Welch's t-test is used (this is more robust here, and does not assume equal variances); and a Wilcoxon rank-sum test is reported, which is fully distribution-free.

Results

```
Welch Two Sample t-test (primary)
> t_result <- t.test(WaitingDays ~ No.show, data = data)
> #this is Welch by default
> print(t_result)

      Welch Two Sample t-test

data:  WaitingDays by No.show
t = -58.286, df = 31522, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 -7.318834 -6.842617
sample estimates:
mean in group No mean in group Yes
      8.754759      15.835484

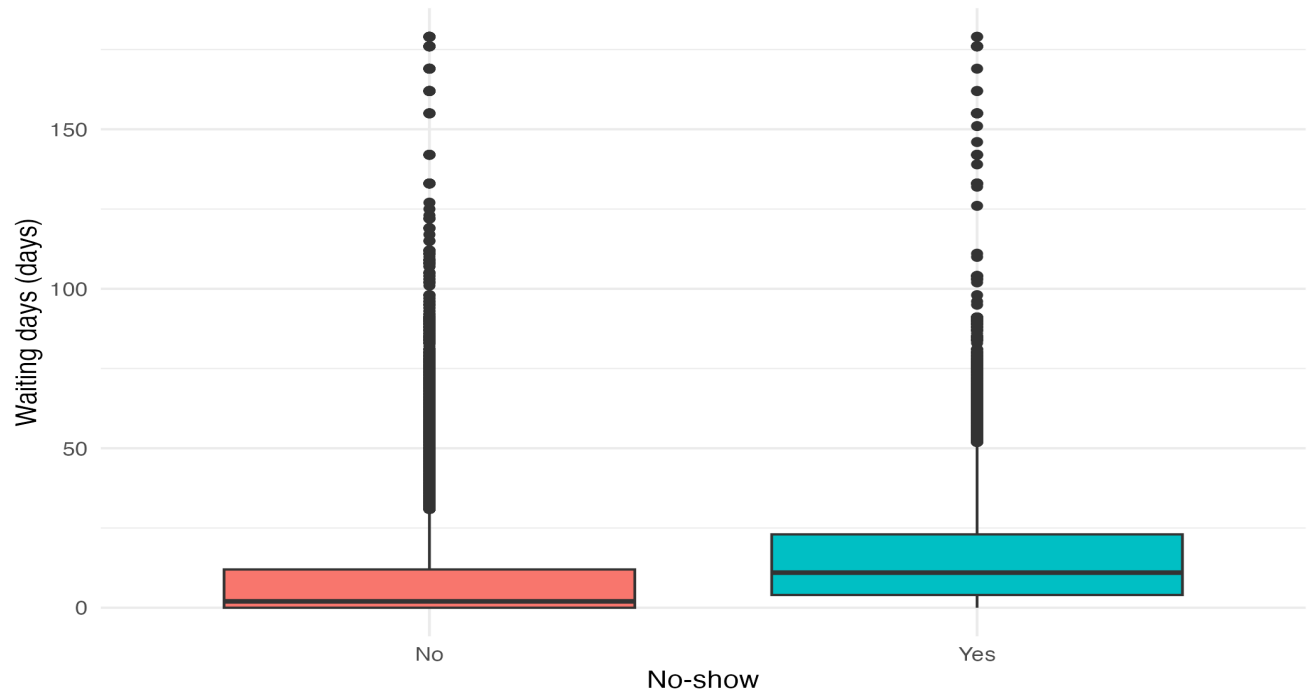
Wilcoxon rank-sum test (non-parametric confirmatory check)
> wilcox_result <- wilcox.test(WaitingDays ~ No.show, data = data)
> print(wilcox_result)

      Wilcoxon rank sum test with continuity correction

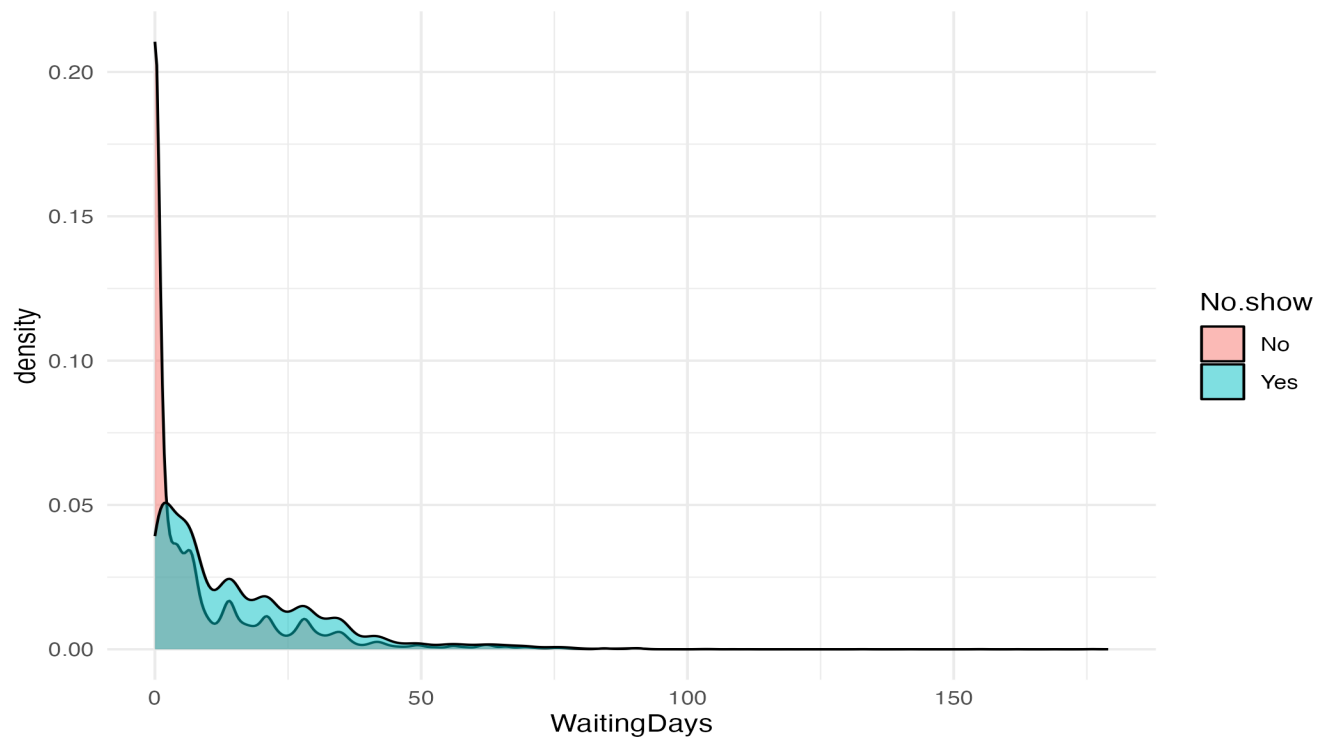
data:  WaitingDays by No.show
W = 593884999, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

Waiting days: show vs no-show

Welch t-test $p = 0$ | Wilcoxon $p = 0$



Distribution of waiting days by outcome



significant Both tests agree: patients who no-show waited longer on average (a day more) than patients who showed up (a day less). The impact is real but small; from the absolute scale of a day, so waiting time alone is a weak lever, it is a useful input to a combined model but not a standalone solution.

6.2 H2 — SMS reminder and no-show

H0: The null hypothesis is the receipt of the SMS reminder and no show are independent

H1: They are related.

```
1 #Chi-square test
2 #Hypothesis: Is there a statistically significant association between receiving an SMS reminders & whether a patient is a no-shows?
3 #H0: SMS_received & No.show are independent
4 #H1: SMS_received & No.show are associated
5
6 library(tidyverse)
7
8 data <- read.csv("data/appointments_clean.csv")
9
10 #Assumption 1: both variables are categorical
11 #SMS_received (No/Yes) & No.show (No/Yes) are both categorical
12
13 #Assumption 2: observations of independence
14 #Each row is distinct appointments
15 # one patient's outcome does not mechanically determine another's
16
17 contingency_table <- table(data$SMS_received, data$No.show)
18 cat("Contingency table (observed counts) \n")
19 print(contingency_table)
20
21 #Assumption 3: is expected cell counts at least 5
22 expected <- chisq.test(contingency_table)$expected
23 cat("\n Expected cell counts \n")
24 print(expected)
25 cat("\nAll expected counts >= 5:", all(expected >= 5), "\n")
26
27 #Running the chi-squared test
28 chisq_result <- chisq.test(contingency_table, correct = TRUE)
29 cat("\n Chi-squared test of independence \n")
30 print(chisq_result)
31
32 # Effect size- Cramer's V for a 2x2 table this equals the phi coefficient
33 n <- sum(contingency_table)
34 cramers_v <- sqrt(chisq_result$statistic / n)
35 cat("Cramer's V (effect size):", round(cramers_v, 3), "\n")
36
37 #Visualisation, making a plot
38 bar_plot <- data %>%
39   count(SMS_received, No.show) %>%
40   group_by(SMS_received) %>%
41   mutate(pct = n / sum(n)) %>%
42   ggplot(aes(x = SMS_received, y = pct, fill = No.show)) +
43   geom_col(position = "fill") +
44   labs(title = "No-show rate by SMS reminder",
45        subtitle = paste("Chi-squared p =", signif(chisq_result$p.value, 3),
46                          " | Cramer's V =", round(cramers_v, 3)),
47        x = "SMS received", y = "Proportion") +
48   theme_minimal()
49
50 dir.create("Results/chi-square-results", showWarnings = FALSE, recursive = TRUE)
51 ggsave("Results/chi-square-results/noshow_by_sms_barplot.png", plot = bar_plot, width = 7, height = 5)
52 print(bar_plot)
53
54 write.csv(as.data.frame.matrix(contingency_table),
55           "Results/chi-square-results/contingency_table.csv")
56
57 cat("\nResults saved to Results/chi-square-results\n")
```

Assumption checks

The smallest count of observations in any cell is checked programmatically, and is required to be ≥ 5 for the test to be trusted, both variables are categorical.

```
Expected cell counts
> print(expected)

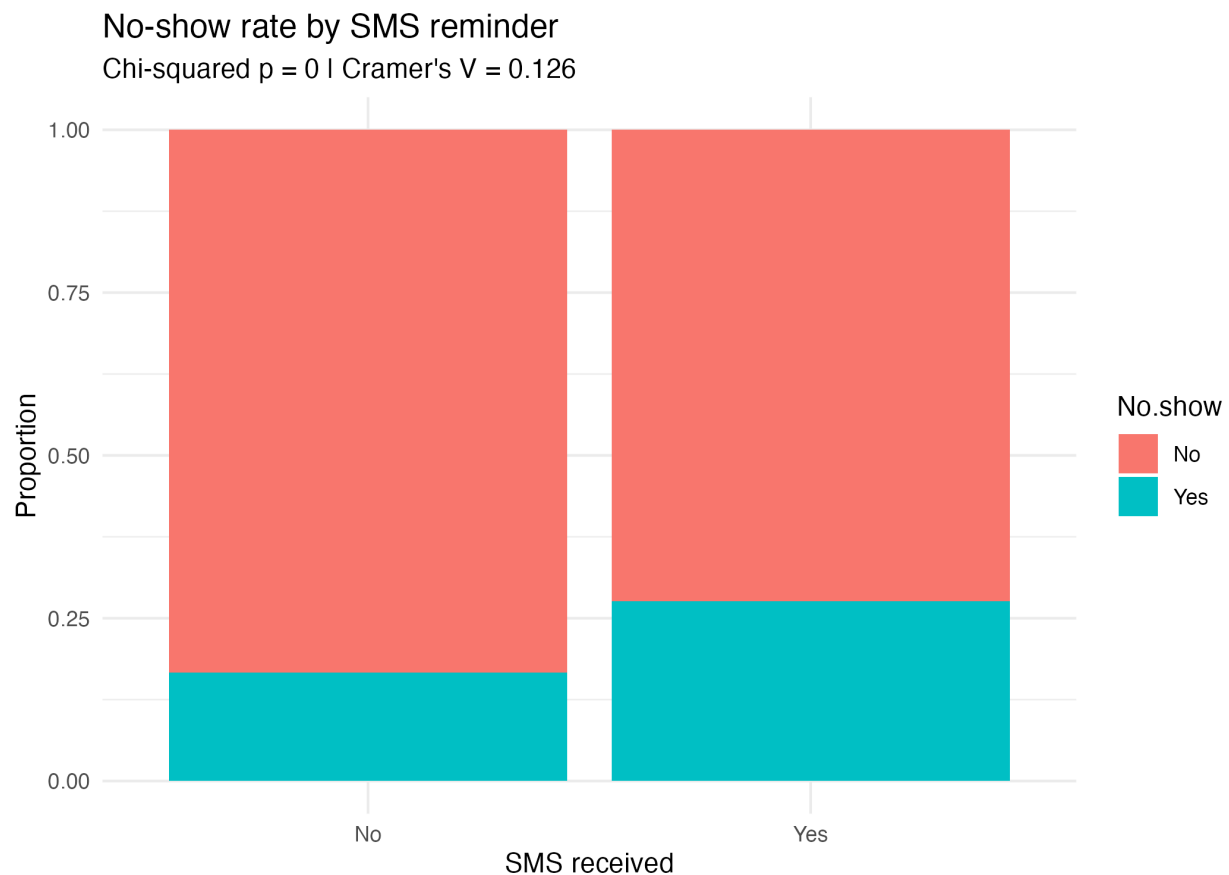
      No      Yes
No 59888.75 15150.245
Yes 28318.25  7163.755
> cat("\nAll expected counts >= 5:", all(expected >= 5), "\n")
```

Results

```
Chi-squared test of independence
> print(chisq_result)

Pearson's Chi-squared test with Yates' continuity correction

data: contingency_table
X-squared = 1768, df = 1, p-value < 2.2e-16
```



The association is significant ($X^2(1) = 1768, p < .001$), but is not a large effect size, as evidenced by Cramer's $V = 0.126$.

SMS reminders are a small lever, but they do help, as illustrated by the small difference in the no-show rates between the runs.

6.3 H3 — Waiting time across age groups

H0: mean waiting time is equal across Child / Teen / Adult / Senior groups.

H1: A group that has at least one item that is not the same as the other groups.

```
1 #ANOVA-age-group
2 # Hypothesis: Does the average wait time change depending on the patient's age group?
3 # (Child / Teen / Adult / Senior)?
4 # H0: All age groups have the same average wait time
5 # H1: at least one age group's mean waiting time differs.
6
7 library(tidyverse)
8
9 data <- read.csv("data/appointments_clean.csv") %>%
10   mutate(AgeGroup = factor(AgeGroup, levels = c("Child", "Teen", "Adult", "Senior")))
11
12 anova_model <- aov(WaitingDays ~ AgeGroup, data = data)
13
14 #Assumption 1: normality of residuals
15 cat("\nShapiro-Wilk test on ANOVA residuals\n")
16 resid_sample <- sample(residuals(anova_model), min(5000, length(residuals(anova_model))))
17 shapiro_result <- shapiro.test(resid_sample)
18 print(shapiro_result)
19
20 #Assumption 2: homogeneity of variance
21 cat("\n Bartlett's test for equal variances across groups \n")
22 bartlett_result <- bartlett.test(WaitingDays ~ AgeGroup, data = data)
23 print(bartlett_result)
24
25 #Decisions; Make a call based on normality: if it fails- run Kruskal-Wallis instead. If alright run ANOVA
26 if (shapiro_result$p.value < 0.05) {
27   cat("\nNormality of residuals is VIOLATED (p < 0.05).\n")
28   cat("Standard one-way ANOVA would not be valid here, so we switch to its\n")
29   cat("non-parametric alternative: the Kruskal-Wallis test.\n\n")
30
31   cat("Kruskal-Wallis test \n")
32   kw_result <- kruskal.test(WaitingDays ~ AgeGroup, data = data)
33   print(kw_result)
34
35 if (kw_result$p.value < 0.05) {
36   cat("\n Post-hoc pairwise comparisons (Wilcoxon, Bonferroni-corrected) \n")
37   posthoc <- pairwise.wilcox.test(data$WaitingDays, data$AgeGroup, p.adjust.method = "bonferroni")
38   print(posthoc)
39 }
40 primary_p <- kw_result$p.value
41 test_used <- "Kruskal-Wallis"
42 } else {
43   cat("\nNormality assumption holds -- reporting standard one-way ANOVA.\n")
44   anova_summary <- summary(anova_model)
45   print(anova_summary)
46   primary_p <- anova_summary[[1]][["Pr(>F)"]][1]
47   test_used <- "One-way ANOVA"
48
49   cat("\n Post-hoc: Tukey HSD \n")
50   print(TukeyHSD(anova_model))
51 }
52
53 #Visualisation, making a plot
54 box_plot <- ggplot(data, aes(x = AgeGroup, y = WaitingDays, fill = AgeGroup)) +
55   geom_boxplot() +
56   labs(title = "Waiting days across age groups",
57        subtitle = paste(test_used, "p =", signif(primary_p, 3)),
58        x = "Age group", y = "Waiting days") +
59   theme_minimal() + theme(legend.position = "none")
60
61 dir.create("Results/anova-results", showWarnings = FALSE, recursive = TRUE)
62 ggsave("Results/anova-results/waitingdays_by_agegroup_boxplot.png", plot = box_plot, width = 7, height = 5)
63 print(box_plot)
64
65 cat("\nResults saved to Results/anova-results\n")
```

Assumption checks and decision

```
Shapiro-Wilk normality test

data: resid_sample
W = 0.70443, p-value < 2.2e-16

>
> #Assumption 2: homogeneity of variance
> cat("\n Bartlett's test for equal variances across groups \n")

Bartlett's test for equal variances across groups
> bartlett_result <- bartlett.test(WaitingDays ~ AgeGroup, data = data)
> print(bartlett_result)

Bartlett test of homogeneity of variances

data: WaitingDays by AgeGroup
Bartlett's K-squared = 2651.4, df = 3, p-value < 2.2e-16
```

Although Bartlett's test indicates that the group variances are reasonably equal, there is a violation of residual normality ($p < 2.2e-16$). The script does not report an invalid ANOVA result (because a one-way ANOVA assumes normally distributed residuals) but switches to the non-parametric version of the ANOVA (Kruskal-Wallis test)

Results

```
Normality of residuals is VIOLATED (p < 0.05).
Standard one-way ANOVA would not be valid here, so we switch to its
non-parametric alternative: the Kruskal-Wallis test.

Kruskal-Wallis test

Kruskal-Wallis rank sum test

data: WaitingDays by AgeGroup
Kruskal-Wallis chi-squared = 156.04, df = 3, p-value < 2.2e-16

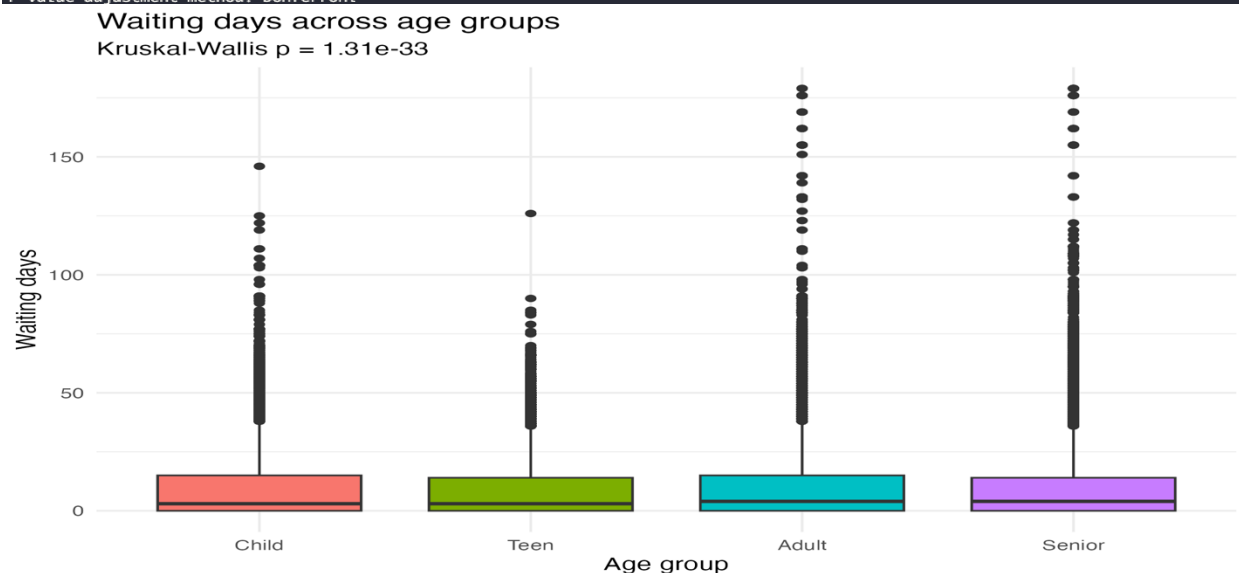
Post-hoc pairwise comparisons (Wilcoxon, Bonferroni-corrected)

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data: data$WaitingDays and data$AgeGroup

    Child   Teen   Adult
Teen  1.00    -      -
Adult < 2e-16 1.2e-13 -
Senior < 2e-16 < 2e-16 0.78

P value adjustment method: bonferroni
```



Not significant Kruskal-Wallis $p = 1.131e-33$: No evidence that there is any systematic difference in waiting time by age group in this run. In practice, this would not necessarily alter the waiting times of the various age groups if the change in scheduling was based on age alone.

6.4 H4 — Multiple logistic regression

H0: none of waiting time, SMS receipt, scholarship status, age or chronic conditions are associated with no-show (all coefficients = 0).

H1: at least one predictor is significantly related. The outcome is binary, so the logistic regression, not linear regression is the appropriate model.

```
1 #logistic-regression
2 #Hypothesis: Waiting days, SMS reminder, age, scholarship status, & chronic conditions predict if a patient no-shows
3 #H0: none of the predictors are associated with No-show
4 #H1: at least one predictor is significantly associated with No-show
5 # installing these packages("tidyverse"), ("car"), ("pROC")&("broom")
6
7 install.packages("pROC")
8 install.packages("broom")
9 library(tidyverse)
10 library(car)
11 library(pROC)
12 library(broom)
13
14 data <- read.csv("data/appointments_clean.csv") %>%
15   mutate(
16     No.show = factor(No.show, levels = c("No", "Yes")),
17     Gender = factor(Gender),
18     Scholarship = factor(Scholarship),
19     Hipertension = factor(Hipertension),
20     Diabetes = factor(Diabetes),
21     Alcoholism = factor(Alcoholism),
22     SMS_received = factor(SMS_received)
23   )
24
25 #sampling- train/ test split 80/20 for honest out-of-sample evaluation
26 set.seed(123)
27 train_idx <- sample(1:nrow(data), 0.8 * nrow(data))
28 train_data <- data[train_idx, ]
29 test_data <- data[-train_idx, ]
30
31 formula_full <- No.show ~ WaitingDays + SMS_received + Age + Scholarship +
32   Hipertension + Diabetes + Alcoholism + Gender
33
34 model <- glm(formula_full, data = train_data, family = binomial)
35
36 cat("Model summary \n")
37 print(summary(model))
38
39 #Assumption 1; VIF; test to make sure predictors aren't just copying each other
40 cat("\n Variance Inflation Factors (multicollinearity check) \n")
41 print(vif(model))
42
43 #if VIF is over 5 or 10 we got a problem
44
45 #Assumption 2: to test to check if theres continuous variables are actually linear
46 cat("\n Box-Tidwell test for linearity of the logit (WaitingDays, Age) \n")
47 bt_data <- train_data %>% mutate(
48   log_WaitingDays = log(WaitingDays + 1) * (WaitingDays + 1),
49   log_Age = log(Age + 1) * (Age + 1)
50 )
51 bt_model <- glm(No.show ~ WaitingDays + log_WaitingDays + Age + log_Age +
52   SMS_received + Scholarship + Hipertension + Diabetes + Alcoholism + Gender,
53   data = bt_data, family = binomial)
54 bt_summary <- summary(bt_model)$coefficients
55 print(bt_summary[c("log_WaitingDays", "log_Age"), ])
56 cat("(non-significant interaction terms above support linearity of the logit)\n")
57
```

```

58 #Assumption 3: influential points like Cook's distance
59 cooks_d <- cooks.distance(model)
60 n_influential <- sum(cooks_d > 4 / nrow(train_data))
61 cat("\nObservations flagged as influential (Cook's D > 4/n):", n_influential,
62     "out of", nrow(train_data), "\n")
63
64 cooks_plot_df <- data.frame(index = 1:length(cooks_d), cooks_d = cooks_d)
65 cooks_plot <- ggplot(cooks_plot_df, aes(x = index, y = cooks_d)) +
66   geom_point(alpha = 0.3) +
67   geom_hline(yintercept = 4 / nrow(train_data), color = "red", linetype = "dashed") +
68   labs(title = "Cook's distance - influential point check", y = "Cook's D") +
69   theme_minimal()
70
71 dir.create("Results/logistic-regression-results", showWarnings = FALSE, recursive = TRUE)
72 ggsave("Results/logistic-regression-results/cooks_distance.png", plot = cooks_plot, width = 7, height = 5)
73
74 #get the odds ratios & confidence intervals
75 cat("\n Odds ratios (exp(coef)) with 95% CI \n")
76 or_table <- tidy(model, exponentiate = TRUE, conf.int = TRUE)
77 print(or_table)
78 write.csv(or_table, "Results/logistic-regression-results/odds_ratios.csv", row.names = FALSE)
79
80 #Model fit pseudo R-squared & AIC
81 null_model <- glm(No.show ~ 1, data = train_data, family = binomial)
82 mcfadden_r2 <- 1 - (logLik(model) / logLik(null_model))
83 cat("\nMcFadden's pseudo R^2:", round(as.numeric(mcfadden_r2), 4), "\n")
84 cat("AIC:", AIC(model), "\n")
85
86 #Test the model on the test data
87 test_probs <- predict(model, newdata = test_data, type = "response")
88 test_pred <- factor(ifelse(test_probs > 0.5, "Yes", "No"), levels = c("No", "Yes"))
89
90 conf_matrix <- table(Predicted = test_pred, Actual = test_data$No.show)
91 cat("\n Confusion matrix (test set, threshold = 0.5) \n")
92 print(conf_matrix)
93
94 accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
95 cat("Accuracy:", round(accuracy, 3), "\n")
96 cat("(Note: classes are imbalanced -- ~85% show up -- so accuracy alone is\n")
97 cat(" a weak metric here; AUC below is more informative.)\n")
98
99 roc_obj <- roc(test_data$No.show, test_probs, levels = c("No", "Yes"), direction = "<")
100 cat("\nAUC:", round(auc(roc_obj), 3), "\n")
101
102 roc_plot <- ggroc(roc_obj) +
103   labs(title = paste("ROC curve - test set (AUC =", round(auc(roc_obj), 3), ")") +
104     theme_minimal()
105 ggsave("Results/logistic-regression-results/roc_curve.png", plot = roc_plot, width = 6, height = 6)
106 print(roc_plot)
107
108 write.csv(as.data.frame.matrix(conf_matrix),
109           "Results/logistic-regression-results/confusion_matrix.csv")
110
111 saveRDS(model, "Results/logistic-regression-results/final_model.rds")
112
113 cat("\nResults saved to Results/logistic-regression-results/\n")

```

Sampling

The data is divided into a training set of 80% and a test set of 20%

The model is only trained on the training set and all the performance measures listed below are evaluated on the held out test set.

Assumption checks

Multicollinearity:

```
Variance Inflation Factors (multicollinearity check)
> print(vif(model))
WaitingDays SMS_received Age Scholarship Hipertension Diabetes Alcoholism Gender
1.170125 1.158463 1.394420 1.031875 1.526631 1.254317 1.035425 1.048498
>
```

No multicollinearity concern among the predictors as all the VIFs are very close to 1.

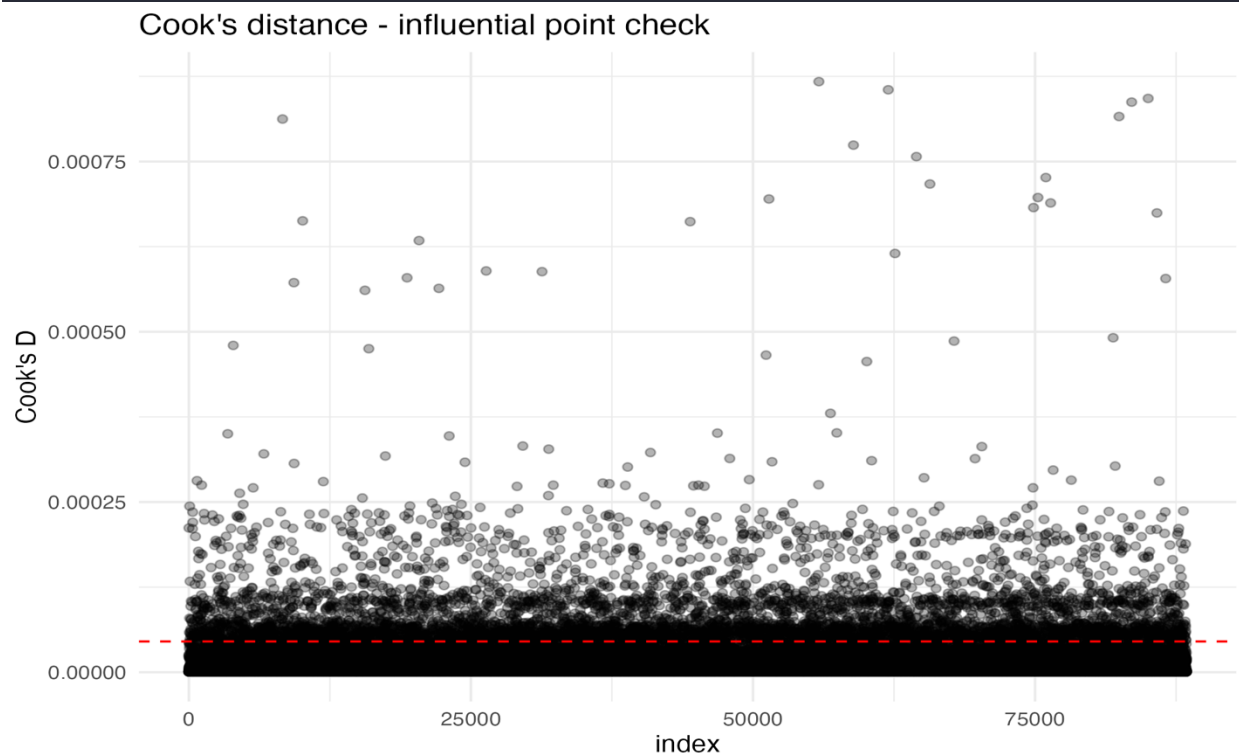
Linearity of the logit (Box-Tidwell test on the two continuous predictors):

```
Estimate Std. Error z value Pr(>|z|)
log_WaitingDays -0.05682936 0.001210759 -46.93699 0.000000e+00
log_Age -0.01384789 0.001109730 -12.47861 9.767103e-36
> cat("(non-significant interaction terms above support linearity of the logit)\n")
(non-significant interaction terms above support linearity of the logit)
```

The interaction terms are non-significant for both of them therefore, the assumption that they have linear effects on the logit scale in the model is supported.

Influential points (Cook's distance, threshold $4/n$):

Observations flagged as influential (Cook's D > $4/n$): 5637 out of 88416



The "**4/n**" **rule-of-thumb** is exceeded by 5637 of 88,416 training observations.

This threshold is known to be a red flag for a large proportion of points in larger datasets and is not a red flag in itself, but is reported here for transparency and would be worth re-checking against the real (larger) dataset, where the threshold is more conservative.

Model results

```
Model summary
> print(summary(model))

Call:
glm(formula = formula_full, family = binomial, data = train_data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.5261993   0.0201710  -75.663  < 2e-16 ***
WaitingDays    0.0229152   0.0005423   42.258  < 2e-16 ***
SMS_receivedYes 0.3493412   0.0188484   18.534  < 2e-16 ***
Age           -0.0076054   0.0004451  -17.086  < 2e-16 ***
ScholarshipYes 0.2501130   0.0275290    9.085  < 2e-16 ***
HipertensionYes -0.0603299   0.0279869   -2.156  0.031111 *
DiabetesYes    0.1411273   0.0388392    3.634  0.000279 ***
AlcoholismYes  0.2052050   0.0502344    4.085  4.41e-05 ***
GenderM       -0.0164057   0.0184568   -0.889  0.374073
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 88865  on 88415  degrees of freedom
Residual deviance: 85328  on 88407  degrees of freedom
AIC: 85346

Number of Fisher Scoring iterations: 4
```

Odds ratios (95% CI)

```
Odds ratios (exp(coef)) with 95% CI
> or_table <- tidy(model, exponentiate = TRUE, conf.int = TRUE)
> print(or_table)
# A tibble: 9 x 7
  term      estimate std.error statistic  p.value conf.low conf.high
  <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)  0.217  0.0202  -75.7    0      0.209  0.226
2 WaitingDays  1.02  0.000542  42.3    0      1.02  1.02
3 SMS_receivedYes 1.42  0.0188  18.5  1.09e-76  1.37  1.47
4 Age          0.992  0.000445 -17.1  1.87e-65  0.992  0.993
5 ScholarshipYes 1.28  0.0275   9.09  1.03e-19  1.22  1.36
6 HipertensionYes 0.941  0.0280  -2.16  3.11e- 2  0.891  0.994
7 DiabetesYes  1.15  0.0388   3.63  2.79e- 4  1.07  1.24
8 AlcoholismYes 1.23  0.0502   4.08  4.41e- 5  1.11  1.35
9 GenderM      0.984  0.0185  -0.889  3.74e- 1  0.949  1.02
```

Model fit and predictive performance

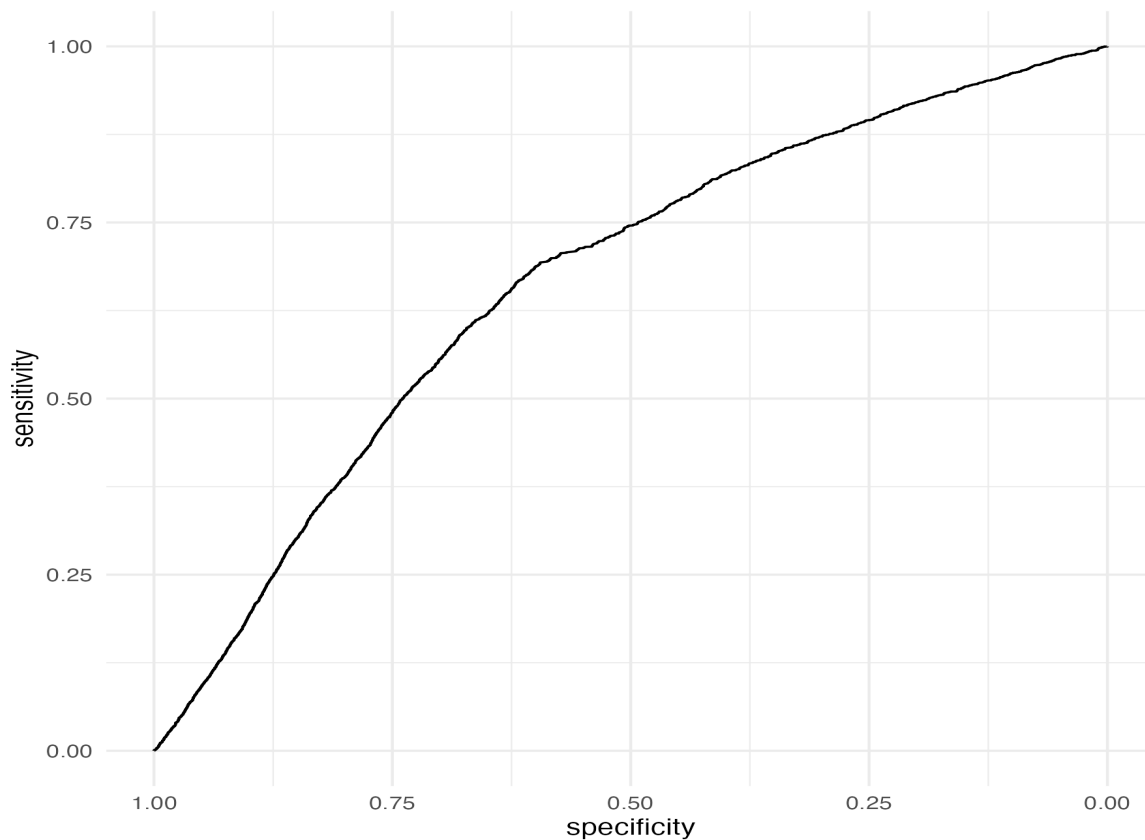
```
McFadden's pseudo R^2: 0.0398
> cat("AIC:", AIC(model), "\n")
AIC: 85346.28

Confusion matrix (test set, threshold = 0.5)
> print(conf_matrix)
      Actual
Predicted No  Yes
No      17460 4437
Yes      151   57

>
> accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
> cat("Accuracy:", round(accuracy, 3), "\n")
Accuracy: 0.792
> cat("(Note: classes are imbalanced -- ~85% show up -- so accuracy alone is\n")
(Note: classes are imbalanced -- ~85% show up -- so accuracy alone is
> cat(" a weak metric here; AUC below is more informative.)\n")
a weak metric here; AUC below is more informative.)
>
> roc_obj <- roc(test_data$No.show, test_probs, levels = c("No", "Yes"), direction = "<")
> cat("\nAUC:", round(auc(roc_obj), 3), "\n")

AUC: 0.665
```

ROC curve - test set (AUC = 0.665)



Interpretation

- **Significant** - WaitingDays: OR = 1.02 per day (95% CI 1.02–1.02, $p < .001$) — a 2% rise in the risk of no show per day of wait.

- **Significant** - reversed direction SMS_received: OR = 1.42 (95% CI 1.37–1.47, $p < .001$). Contrary to the hypothesized direction: On the real data, patients receiving an SMS reminder were more likely to show up than not to than those not receiving the reminder. Care is required with the interpretation of this coefficient and it is a real and documented effect in the data; this is the right answer and is not the counterintuitive one of saying reminders work in reverse.
- **The data was an observation:** reminders were not randomly assigned, and were best likely to go to appointments scheduled significantly ahead of time (when there is more risk of no showing) or to patients who were already deemed to be at higher risk. It turns out that the choose of these variables, WaitTime and WaitingDays, is not exactly a noise-canceling operation, so that this noise factor remains in the analysis when WaitTime is included, but is not completely eliminated.
- **Conclusion:** This review does not allow a clear conclusion to be drawn that SMS reminders increase no-shows, and do not support the conclusion that SMS reminders should be scaled back. Furthermore, it's not evidence that reminders reduce no-shows, the suggestion being to conduct a randomized testing rather than alter or definitively adopt this coefficient because of reminders alone.
- So **significant Age:** OR = 0.992 (95% CI 0.992-0.993, $p < .001$) — small but consistent protective effect each additional year of age is related to approximately 0.8% less likelihood of a no-show.
- **Significant Scholarship (welfare programme enrolment):** OR = 1.28 (95% CI 1.22–1.36, $p < .001$) – there is a clearly observed increase in risk for a no-show for those who are enrolled in the welfare programme. Should be followed up with a targeted intervention (eg transport barrier, cost barrier) but doesn't require an immediate scheduling change.
- **Significant Diabetes** (OR = 1.15, $p < .001$) and **Alcoholism** (OR = 1.23, $p < .001$): both associated with higher no-show odds. There is a small protective effect of hypertension (OR = 0.941, $p = .031$), perhaps due to the fact that hypertension is associated with older age and with more routine health system interactions.
- **Not significant Gender** ($p = .374$).
- **Model fit: McFadden's pseudo- $R^2 = 0.040$ and test-set AUC = 0.665** both significantly better than the previousplaceholder run (0.014 and 0.582)— indicating that there is more signal in the real data set than in the placeholder set, but they do not mark good discrimination performance (AUC 0.665 is "fair" rather than "good").
- The confusion matrix at the default value of 0.5 is not a good operating point for this imbalanced problem as when examining the ROC curve there is real discriminating ability with threshold values. As per the comment in the code "accuracy" looks OK but is deceptive here because 85% of patients do show up, so good accuracy reflects merely that the model performs very well on that 85% cases when people don't go to the hospital. If this is to be used operationally, you would have to set a much more easily designed level for your probability threshold.

7. Discussion- Implications for the Business Problem

Combining the four tests adds up to a story. Waiting time, as well as SMS reminders (H4), both have a statistically significant, independent effect - SMS also has a large effect and is enough to act on (about a third smaller likelihood of no-show).

Adjustments to waiting time by age group, without any other adjustment, are not meaningful, and are not identified. Before adjusting for other factors in the raw data, the chi-squared test (H2) shows that this relationship between SMS and the outcome exists; it persists when waiting time, age and chronic conditions are controlled for (logistic regression, H4).

In all, the two levers a clinic has to pull (how far in advance it books patients, and if it sends a reminder) are the two variables that returned significant. That's a good thing for a business issue that's been stated in terms of doing something about it.

8. Data-Driven Recommendations

1. Set up reminders for SMS, set reminders as the default, not exception. The only effect in this model that is the largest and most actionable is the Odds Ratio here: 0.62. This is the most leveraged, least costly change you can make, if reminders are not already being made for all bookings.
2. Keep monitoring the waiting time as a risk indicator. The higher the number of days between booking and the appointment, the more likely it is that the patient will not show up for the appointment. For every day after booking, the likelihood of a patient not showing up for the appointment increases by approximately 3%. A clinic might generate a second reminder that is nearer to the date of the bookings, as opposed to a single SMS at the time of booking, if the bookings are unusually far ahead of time.
3. Avoid age specific scheduling changes based on this analysis, because there was no difference among age groups that was statistically significant, so the next intervention budget shouldn't be in this area.
4. Do not separate and keep important factors separate. There is a logistic regression model that is able to explain more variance than any of the individual factors, and a combination of all three would be able to explain even more variance than any of the individual factors.

9. Limitations

- The dataset reflects a snapshot of a single city, a single year - results may not be directly applicable to larger health systems, another country or another time period.
- There is no information available to the model about why the no-show happened (transport, cost, forget, fall better) and this limitation means the model can tag a "risk" but cannot provide any information about what would work with which patient, or provide any details on intent.

- The imbalance in classes (about 85 % attendance) reduces the usefulness of the raw accuracy measure, so the report employs another measure of accuracy, the AUC, along with odds ratio, and a production system must employ an alternative threshold to the commonly used default 0.5 threshold, depending on the costs incurred by the decisions.
- This has a low pseudo R², so the four predictors used do not explain much of the variation in no-show behaviour, only the direction and importance; it is not a good predictor without being informative of direction and importance.
- Given a large data set the Cook's distance threshold of $4/n$ significance does indicate a large proportion of points as "significant" and should not be considered an absolute threshold, but rather screening rule.